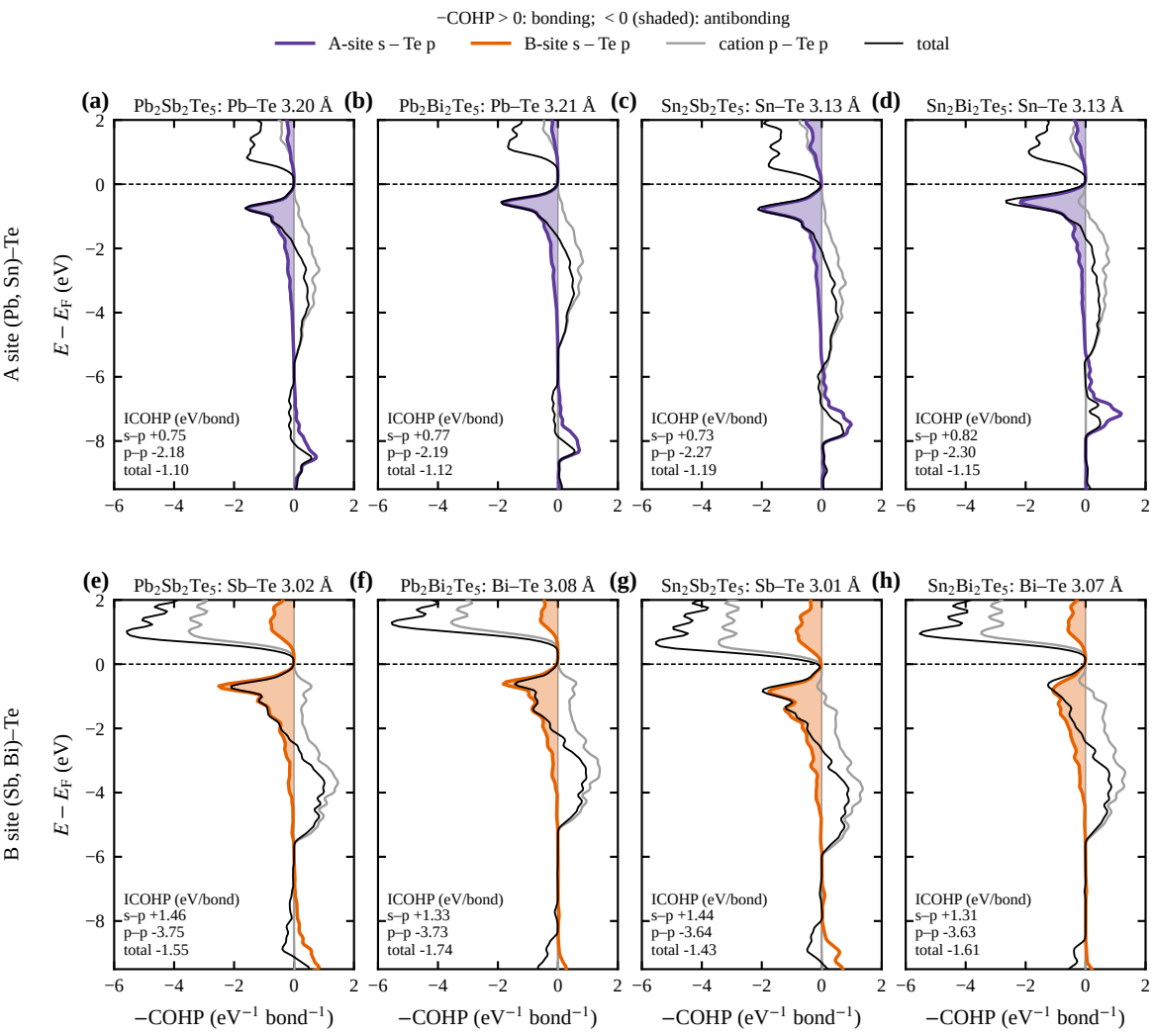

−COHP > 0: bonding; < 0 (shaded): antibonding
A-site s – Te p
B-site s – Te p
cation p – Te p
total
(a) Pb2Sb2Te5: Pb–Te 3.20 Å
(b) Pb2Bi2Te5: Pb–Te 3.21 Å
(c) Sn2Sb2Te5: Sn–Te 3.13 Å
(d) Sn2Bi2Te5: Sn–Te 3.13 Å
(e) Pb2Sb2Te5: Sb–Te 3.02 Å
(f) Pb2Bi2Te5: Bi–Te 3.08 Å
(g) Sn2Sb2Te5: Sb–Te 3.01 Å
(h) Sn2Bi2Te5: Bi–Te 3.07 Å
ICOHP (eV/bond)
s–p +0.75
p–p -2.18
total -1.10
ICOHP (eV/bond)
s–p +0.77
p–p -2.19
total -1.12
ICOHP (eV/bond)
s–p +0.73
p–p -2.27
total -1.19
ICOHP (eV/bond)
s–p +0.82
p–p -2.30
total -1.15
ICOHP (eV/bond)
s–p +1.46
p–p -3.75
total -1.55
ICOHP (eV/bond)
s–p +1.33
p–p -3.73
total -1.74
ICOHP (eV/bond)
s–p +1.44
p–p -3.64
total -1.43
ICOHP (eV/bond)
s–p +1.31
p–p -3.63
total -1.61
A site (Pb, Sn)–Te
B site (Sb, Bi)–Te
E − EF (eV)
−COHP (eV−1 bond−1)